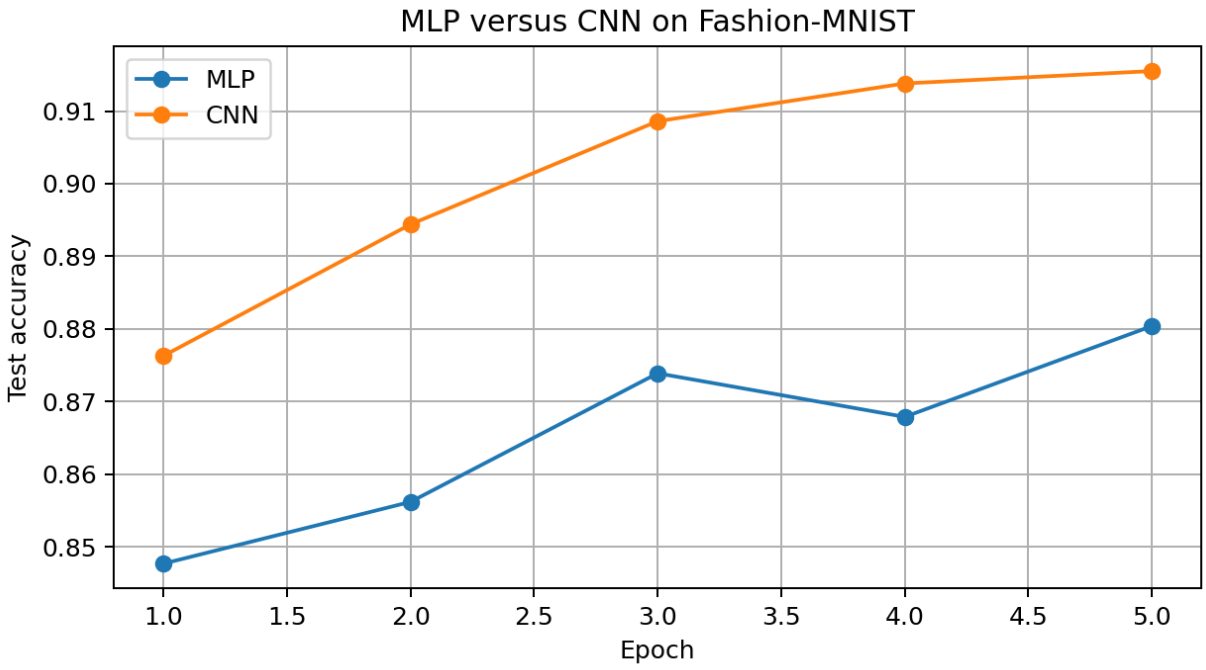
Week 3 Experiment Report

CNN classification, Kaggle workflow, and PCA + K-means

Part 1. CNN classification on Fashion-MNIST

I trained the required CNN for five epochs on Fashion-MNIST. The network uses two convolution blocks followed by a fully connected classifier. The comparison uses the same dataset, batch size, optimizer, learning rate, and number of epochs as the Week 2 MLP run.

Item	Setting
Network	Conv2d(1,32,3,padding=1) → ReLU → MaxPool(2) → Conv2d(32,64,3,padding=1) → ReLU → MaxPoo
Optimizer	Adam
Learning rate	0.001
Batch size	128
Epochs	5
Final training loss	0.1905
Final test accuracy	91.55%



The measured comparison is:

Model	Test Accuracy
MLP	88.04%
CNN	91.55%

The CNN performed better in this run. It reached 91.55% test accuracy, compared with 88.04% for the MLP.

Part 2. Kaggle Digit Recognizer

The notebook loads Kaggle train.csv/test.csv, trains a CNN with an internal validation split, and writes a submission CSV. The local workflow was rerun after the successful platform submission.

Item	Record
Network	The same two-convolution CNN used in Part 1
Optimizer	Adam
Learning rate	0.001
Batch size	128
Epochs	5
Validation accuracy	0.98488 (fresh local rerun)
Kaggle platform score	0.98353 (submitted successfully)

The submitted Kaggle file received 0.98353. A separate local rerun reached 0.98488 validation accuracy. The local number is not claimed to be from the exact model used for the submitted file.

Required submission format: ImageId,Label. ImageId starts at 1.

Part 3. PCA + K-means

I used 10,000 randomly selected Fashion-MNIST training images for a controlled and tractable clustering experiment. Each image was flattened to 784 dimensions. K-means used K=10, n_init=10, and random_state=42. NMI and ARI were calculated only after clustering, using the labels for evaluation.

Representation	Dimension	NMI	ARI
Raw pixels	784	0.5113	0.3456
PCA-50	50	0.5294	0.3823
PCA-20	20	0.5280	0.3805
PCA-10	10	0.5087	0.3441

PCA-50 and PCA-20 were slightly better than raw pixels in this run. PCA-10 did not improve the scores, which suggests that reducing the dimension too much can remove useful information.

The full code and CSV export are in week3_clustering.ipynb.

Week 3 Questions

1. Which model works better? In this experiment, the CNN worked better than the MLP: 91.55% versus 88.04% test accuracy.

2. Why can convolution help with image data? Convolution looks at local neighborhoods and reuses the same weights across the image, so it preserves spatial structure while using fewer parameters than a fully connected layer.

3. What does a convolutional filter learn? A filter learns a small pattern detector, such as an edge, texture, or more complex local shape.

4. What is a feature map? A feature map is the output produced when one filter scans across an image; its values show where the learned pattern appears.

5. What is the difference between classification and clustering? Classification learns from known labels, while clustering groups samples without labels.

6. Why does clustering not use labels during training? Clustering is unsupervised, so it discovers groups from the input features. Labels are used afterward only to evaluate the groups.

7. What does K mean in K-means? K is the number of clusters the algorithm is asked to find.

8. What is a cluster center? It is the mean feature vector of the samples currently assigned to one cluster.

9. What does PCA do? PCA projects the data onto a smaller set of directions that explain large amounts of variance.

10. Why might raw pixels be poor clustering features? Raw pixels include redundancy and are sensitive to small visual changes, so Euclidean distance may not match semantic similarity well.

11. Why can dimensionality reduction help? It can remove redundancy and noise, reduce computation, and make the main structure easier for K-means to use.

12. Why does cluster ID 0 not need to represent class 0? Cluster IDs are arbitrary names. K-means can label the same grouping with different numbers without changing the result.